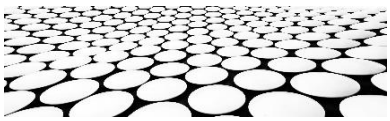
CSI5180 Topics in AI
Virtual Assistants



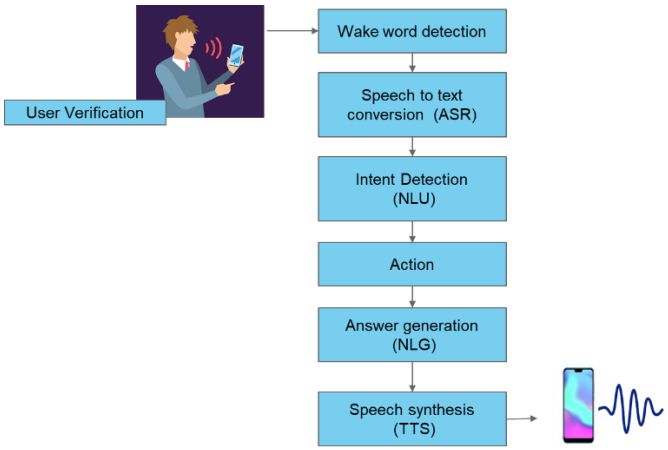
Voice Assistant Project

Overview + Step 1 requirements

GOAL

The goal of the programming project is to build your own Voice Assistant which will respond to a defined set of intents. **For this project, students are asked to work in groups of 3.**

The project will in part revisit modules explored during the activities (code from those can be reused), and in part require the development of new modules. The project aims at a full integration of the modules (as shown in the Figure below) towards a Voice Assistant.



The project plays an important part in our course, and I therefore expect that each group will put effort into developing a fully functional VA.

SUBMISSION DEADLINES

There are three milestones in the project, each with its deadline. The milestones are mentioned below, and your focus now is on your Project Definition. The project is worth 32%, split as below.

(1) Project Definition (10%)

- *Deadline Thursday March 12.*
- Submit a small report to describe your VA (ideas for implementation of different modules, choice of specialized domain API, choice of control system).
- The professor will review your document and provide feedback (approval or modification). You might be asked for clarifications/modifications before you can proceed with your project.

(2) Project Pitch / Demos / Peer reviews (20%)

- During class of April 14th (last class), each group will do a short presentation (project pitch) and a demo. Your presence is mandatory.
- Each group will also be required to perform peer review on a few other groups.
- The night before, submit the slides for your project pitch.
- Your VA must be fully functional at that time to demo to your colleagues.

(3) Final Video Report (70%)

- Deadline, April 17th
- Submit a 10-minute video presentation (12 minutes max). The presentation should describe each module and include a demonstration of your software.

PLEASE NOTE: The focus of the present document is on Project Definition. The requirements for the pitch presentation, demos, video report will be provided on Brightspace later in March.

REQUIREMENTS

Your Voice Assistant must contain the following modules:

1. User verification (New module, using Activity 1 dataset)
2. Wake Word Detection (Module inspired from Activity 1)
3. Automatic Speech Recognition (Use of available software - Whisper)
4. Intent Detection (Module inspired from Activity 2)
5. Fulfillment (Module inspired from Activity 3)
6. Answer Generation (Module inspired from Activity 3)
7. Text-To-Speech (Module inspired from Activity 4, use of available software)

Your VA should have these steps available in a pipeline. The VA should also allow a "by-pass" at each step, so a user can skip a step when it is not successful.

Your VA should have a minimal UI (could be more complex UI if you desire) to be able to execute/skip the different steps as well as display some information related to its tasks.

Let's go through the different modules. You will need to provide ideas for modules 1 to 6 in your Project Definition.

1. <u>User verification/identification</u>

The purpose of this module is to identify users of the system from their voice. You can limit this to the members of your team or make it more general.

For doing so, you can use the data gathered during Activity 1 which contains the voice of the different students in the class.

You can treat this as User verification task (user not in the training set) or User Identification task (only recognize people who are in the training set). In either case, the task should be a text-independent task, meaning that the user can speak any word they want.

To help you develop this module, you should go back to the slides from Module 2 on User Verification.

- *Input: User's voice.*
- *Task: Determine if user is who they say they are.*
- *Input State: VA is "locked".*
- *Output State: Once the User is verified or identified, the system goes into a "Unlocked" state waiting to be awaken for further instructions. If not, the system stays in "Locked" state.*
- *By-pass: Allow the user to enter a code to unlock the system.*

2. Wake Word Detection

The purpose of this module is to wake up the system. The previous step was unlocking it, not waking it up. You have developed a module with the Wake Word "Hey Atlas". You can adapt that module.

- *Input: User's voice.*
- *Task: Classify the audio into Wake Word or not.*
- *Input State: VA is "Unlocked", but in "Sleep" mode.*
- *Output State: If Wake Word is recognized, show "Awake" state. If not, show "Sleep" state.*
- *By-pass: Allow the user to type the Wake Word (instead of voice).*

3. Automatic Speech Recognition

The purpose of this module is to transcribe the spoken command from the user. You can make this unilingual (simpler case) or multilingual (more challenging - optional). In the multilingual case, you should output the transcribed speech in both the source language and in English (as the next module will continue with English only).

- *Input: User's voice.*
- *Task: Transcribe the sentence spoken.*
- *Output: Transcribed Speech shown on screen. Await confirmation.*
- *By-pass: Allow the user to correct the transcription.*

4. Intent Detection

The purpose of this module is to classify the input sentence toward one of the predefined intents, as well as recognize the parameters (slot filling). *This module needs to be developed in relation to the next module on Fulfillment. What your VA recognizes as intents is what it will be able to fulfill.*

During Activity 2, we have developed a joint intent detection and slot classification module doing fine-tuning of a pre-trained BERT model. You can adapt the code.

Basic intents

There is a basic set of mandatory intents that your VA must recognize.

a) Out-Of-Scope (OOS): "Do my grocery shopping"
b) Greetings: "Hello"
c) Goodbye: "Thank you"
d) Timer: "Set a soup timer for 2 minutes"
e) Weather: "Does it rain in Paris today?"

Then, where your VA becomes unique is through 2 additional sets of intents that it will recognize.

Specialized Domain Intents

You must choose one free API for a domain of your choice (e.g. music, movies, finance, art, environment, etc.) in which your VA will specialize.

- Explore the API to see what it can do. The same API usually allows for different requests. Based on the possible requests, define 5-10 intents for that domain.
- This component is *stateless* in the sense that each query is independent of previous interactions.

Interactive System Control

Choose one API or provide a simulated environment (e.g. game, smart home, etc) where the VA will control the environment.

- Implement a set of intents (5-10) to interact with the system (e.g. turn-right or jump for a game character; set the thermostat to 20C or turn off the bedroom lamp for a home system). This component is *stateful* as the VA must remember changes across interactions.

What will make your VA unique among all the VAs developed by your colleagues is both your choice of Specialized Domain as well as your choice of control system.

- *Input: Sentence (text transcribed from ASR module)*
- *Task: Perform intent classification and slot filling.*
- *Output: Recognized Intent and required Slots.*
- *By-pass: Allow the user to manually enter Intent/Slots. Provide a list to choose from.*

PLEASE NOTE: You are not required to implement a dialog system to obtain missing information for mandatory slots. For example, if the user says "set timer", the intent is incomplete since the actual time is required. A real system would ask "for how long" engaging in a short dialog. If you wish (optional) you can include this feature. If that feature is not implemented, then the system will not be able to continue to fulfillment and should inform the user.

5. Fulfillment

The purpose of this module is to fulfill the intent.

In Activity 3, we will work on fulfilling intents, so you will have some code to work with and expand.

- *Input: Intent + required Slots.*
- *Task: Transform input into a request to an API and obtain the response. For the control system, there will be no actual response but a change of state.*
- *Output: Answer from API (probably in json format).*
- *By-pass: Have a pre-canned API output.*

6. Answer generation

The purpose of this module is to transform the json answer from the API into Natural Language (textual answer), and to display the result of the control system (visual answer).

In Activity 3, we will work on template generation of answers. You will have code from the Activity to inspire you as a first answer generation approach. In addition, you will be required to explore answer generation using Large Language Models, creating prompts from the json answers to test prompt-based text generation. The theory for this module will be presented but you will need to explore the coding.

Furthermore, for your control system, you will need to visually show changes of state (e.g. change of direction of a character, change of color of a simulated lamp, etc). We will discuss in class how some systems further verbalized confirmation (e.g. bedroom lamped turned off) on top of doing the action. You can decide what style of VA you build and whether it will be verbose or not.

- *Input: Json reply from API (or state change)*
- *Task: Transform the received Json into Natural Language. Or modify the visual display based on a new state.*
- *Output State: Natural Language or Visual Display.*
- *By-pass: Have a pre-canned sentence for the NL.*

7. Text-to-Speech

The purpose of this module is to produce an audio signal from the answer sentence from the previous module (for the APIs). Further requirements will be provided for this step. At this point, we have not seen the related necessary material.

- *Input: Generated response in NL*
- *Task: Transform the text into speech.*
- *Output State: Speech*
- *By-pass: no need (we are at the last step)*

SUBMISSION CONTENT

Your short proposal report (max 3 pages) should be submitted in Brightspace. It will be marked on 10 points.

It's a group submission, so you need to first register your group (Project group) and then submit.

Your project definition should provide enough information to show you are on a good path to success.

1. Group information:
    a. Names of the 3 students part of the group. **Your group should be of 3 students.**
2. Short description of the following modules:
    a. User verification/identification: How will you address this problem? What is your training set, task, anticipated output.
    b. Wake Word Detection: Any refinement you can do to the module you built during the activity?
    c. Speech Recognition: Will you work on unilingual, multilingual? Will you continue with Whisper as package or use a different one?
3. Description of your **specialized domain**:
    a. What is your domain of interest?
    b. What API will you be using?
    c. What possible intents can be fulfilled by this API? (mention a few if too many)
    d. Provide 3 examples with:
        i. What is the intent? What are the slots?
        ii. The intent/slots rewritten to obtain information from the API
        iii. The response that will be received from the API
        iv. The rewritten response you wish to output in Natural Language for your user.
4. Description of your **control system**:
    a. What will your VA control?
    b. What API will you be using, or will you simulate the control?
    c. What possible intents can be fulfilled by this controller?
    d. How will you show the state of the system and its changes? Provide a draft of the display.
    e. Provide 3 examples with:
        i. What is the intent? What are the slots?
        ii. How will that change the state of the system?
        iii. How will you display the answer visually?
5. List of references (sources for platforms, APIs, etc).